

Cyber Bullying Text Classifier

Project Synopsis

Mini Project Evaluation -Summer Internship

BCAMJ23515

**BACHELOR OF COMPUTER APPLICATION –
(Cloud Technology & Information Security)**

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1 Project Title

“Cyberbullying Text Classifier: Detecting Toxic Online Comments Using Machine Learning”

2 Domain

Machine Learning / Text Classification

3 Problem Statement

With the rapid growth of social media and online communication platforms, cyberbullying has become a major concern affecting individuals emotionally and mentally. Harmful comments such as threats, insults, and abusive language are often difficult to monitor manually due to the massive volume of online interactions. To address this issue, automated detection of toxic comments using Machine Learning is essential. This project focuses on building a cyberbullying text classifier that can identify and categorize toxic comments effectively. The objective is to assist in creating safer digital spaces by reducing harmful interactions and supporting early detection of cyberbullying.

4 Project Description

The proposed project, *Cyberbullying Text Classifier*, is designed to automatically detect and classify toxic comments from social media text using Machine Learning techniques. The project demonstrates how technology can be used to reduce harmful interactions and support safe online environments.

4.1 Scope of the Work

This project aims to develop a *Cyberbullying Text Classifier* that can automatically detect toxic comments using Machine Learning models. Initially, the scope is limited to preprocessing textual data, extracting features through TF-IDF vectorization, and training models such as Logistic Regression and Support Vector Machine (SVM) to classify comments as toxic or nontoxic.

To enhance usability, the project will be extended into a **web application** where users can input their own comments for classification. In addition, **Business Intelligence (BI) dashboards** will be integrated to visualize patterns, trends, and statistics of toxic comments. The project will also

provide an **informational module** containing awareness content about cyberbullying and its impacts.

4.2 Project Modules

The project structure is divided into the following modules:

1. **Data Preprocessing** – Cleaning text by converting to lowercase, removing punctuation, special characters, and handling emojis.
2. **Feature Engineering** – Applying TF-IDF vectorization to extract meaningful features from text.
3. **Model Training** – Training two models: Logistic Regression and Support Vector Machine (SVM).
4. **Evaluation** – Measuring performance using Precision, Recall, F1-score, and Confusion Matrix.
5. **Result Visualization** – Comparing models through bar charts and confusion matrices.

4.3 High Level Design :

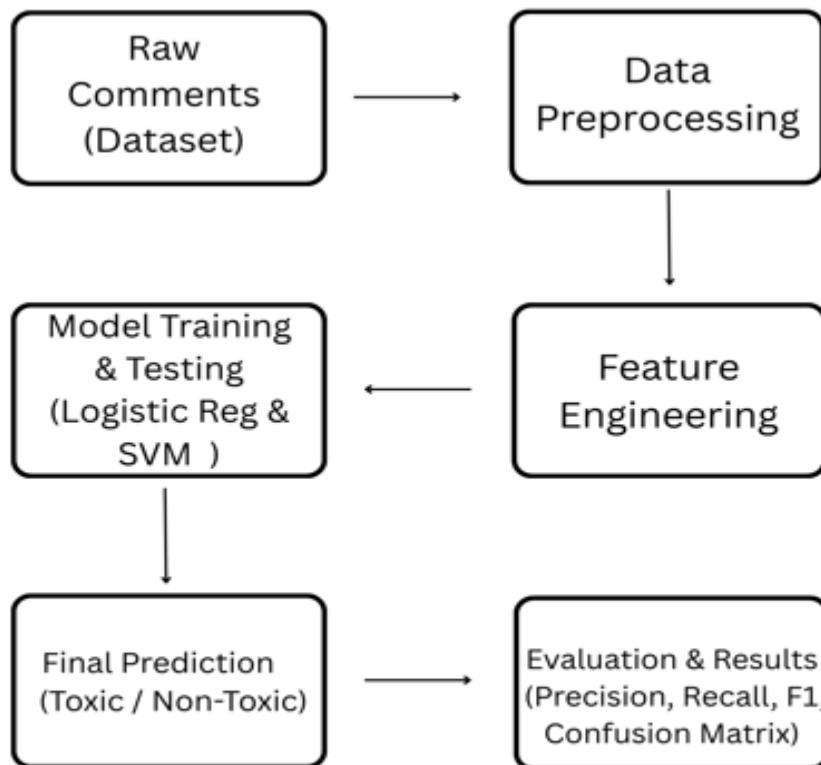


Figure 4.3 : High-Level Workflow of Toxic Comment Detection using Machine Learning.

5. Implementation Methodology

The proposed software system for **Toxic Comment Detection** is designed to automatically identify and classify online comments as *toxic* or *non-toxic*. The methodology involves several sequential phases — from data preprocessing to model evaluation — to ensure efficient and accurate detection.

System Workflow :

The project follows a **pipeline-based workflow**:

1. Input Data

- ❖ User comments (from the Jigsaw dataset).

2. Data Preprocessing

- ❖ Lowercasing text.
- ❖ Removing punctuation, special characters, and extra spaces.
- ❖ Handling emojis (converting them into textual meaning).
- ❖ Retaining meaningful numbers.

3. Feature Engineering

- ❖ Applying **TF-IDF vectorization** to represent comments numerically while highlighting rare but significant toxic words.

4. Model Training

- ❖ Training two machine learning models:
 - **Logistic Regression** (focus on Recall).
 - **Support Vector Machine (SVM)** (focus on Precision & Accuracy).
- ❖ Using **train-test split (80–20, stratified)** to handle class imbalance.

5. Evaluation & Results

- ❖ Metrics: Precision, Recall, F1-score, Accuracy.
- ❖ Visualizations: Confusion Matrix and Bar Chart Comparisons.
- ❖ **Interpretation:** Logistic Regression catches more toxic cases (high Recall), SVM reduces false alarms (high Precision).

6. Output

- ❖ Final classification: **Toxic / Non-Toxic**.

6. Technologies to be used

a. Software Platform

a). Front-end

- ❖ Not applicable (this project is research/ML focused).
- ❖ Jupyter Notebook (interactive interface for development and visualization).

b). Back-end

- ❖ **Programming Language:** Python 3.10

- ❖ **Libraries & Frameworks:**

- NumPy, Pandas → Data handling and preprocessing
- Matplotlib, Seaborn → Visualization of results and metrics
- Scikit-learn → Machine Learning models and evaluation metrics
- Regex → Text preprocessing

b. Hardware Platform

- ❖ **Processor:** Ryzen5 / intel5
- ❖ **RAM:** Minimum 8 GB
- ❖ **SSD:** At least 500 GB (Dataset ~160 K, but processing requires more space)
- ❖ **Operating System:** Windows 11
- ❖ **Editor/IDE:** Jupyter Notebook
- ❖ **Browser:** Google Chrome (for Jupyter interface and visualization)

c. Tools

Tool Name	Developer	Version	Purpose
Jupyter Notebook	Project Jupyter	Latest	Development & Testing
Python	Python Software Foundation	3.10	Programming Language
Scikit-learn	Open Source (BSD License)	1.3+	Machine Learning models & evaluation

Tool Name	Developer	Version	Purpose
Pandas	Open Source (Community)	2.x	Data handling
Matplotlib	Open Source (Community)	3.7+	Plotting graphs, visualization
Seaborn	Open Source (Community)	0.12+	Statistical data visualization
Git/GitHub	GitHub Inc.	Latest	Version control, collaboration

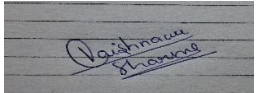
7. Advantages of this Project

1. **Early Detection of Cyberbullying:** Helps in identifying toxic and harmful comments at an early stage.
2. **Safer Online Platforms:** Assists social media platforms, forums, and websites in creating a healthy online environment.
3. **Automated Process:** Reduces the need for manual monitoring of comments, saving time and effort.
4. **Awareness Creation:** Helps spread awareness about cyberbullying and its detection using technology.
5. **Scalable Solution:** Can be applied to large volumes of comments or messages in real time.
6. **Customizable:** Models can be retrained or improved with new data to adapt to changing online behaviour.
7. **Beneficiaries:**
 - ❖ **Online Platforms / Communities** – For moderating harmful content.
 - ❖ **Parents / Teachers** – To monitor and protect students from bullying.
 - ❖ **Researchers / Developers** – For further improvement in AI-based moderation tools.

8. Future Scope and further enhancement of the Project

- 1) The project will be extended into a **web application** where users can input their own comments for real-time classification as toxic or non-toxic.
- 2) **Business Intelligence (BI) dashboards** will be integrated to visualize patterns, trends, and statistics of toxic comments for deeper insights.
- 3) An **informational module** will be added to spread awareness about cyberbullying, its effects, and ways to deal with it.
- 4) The system can be further enhanced with advanced models (e.g., deep learning) to improve accuracy and handle multiple languages.
- 5) The solution can be deployed on **social media platforms, schools, and communities** to promote safe online interactions.

9. Team Details

Project Name	Course Name	Student ID	Student Name	Role	Signature
Cyberbullying Text Classifier	BCA – (CTIS)	TCA2356117	Vaishnawi Sharma	Developer (including implementation, documentation)	

10. Conclusion

This project successfully demonstrates the use of **Machine Learning techniques** for detecting toxic and cyberbullying comments. The approach integrates **data preprocessing, TF-IDF based feature engineering, and classification models (Logistic Regression and SVM)** to achieve effective results.

The innovative aspect of this project lies in the **focus on recall-oriented detection**, ensuring that more harmful comments are identified, which is critical in combating cyberbullying. The evaluation results highlight that while **SVM provides better precision and overall F1-score**, **Logistic Regression offers higher recall**, making it more suitable for this sensitive application.

The main achievement of the project is the development of a **scalable and automated pipeline** that can process large volumes of comments with minimal human intervention. Additionally,

the inclusion of result visualization (confusion matrix and performance comparison) makes the model more interpretable.

This system stands out by emphasizing **awareness creation and social impact**, showing how technology can assist in building safer online communities. With future enhancements such as **web application deployment, BI dashboards, and awareness modules**, the project has strong potential for real-world application and expansion.

11. References

[GitHub Repository](#) : [\[Vaishnawi-Sharma/cyberbullying-text-classifier: A machine learning project that detects cyberbullying in text data. Built using Logistic Regression and SVM, the model analyzes social media comments to classify them as bullying or non-bullying. Includes performance evaluation with confusion matrices and key metrics \(Precision, Recall, F1-score\) to compare both models.\]](#)